



SHALBY
MedTech

SURGICAL TECHNIQUE



ETERNITY™

DUAL MOBILITY HIP SYSTEM

CoCr-Free Titanium Platform
Vitamin E Bearing Technology
Enhanced Stability

SECURE | EFFICIENT | SHIELDED

ETERNITY™

DUAL MOBILITY CUP SYSTEM

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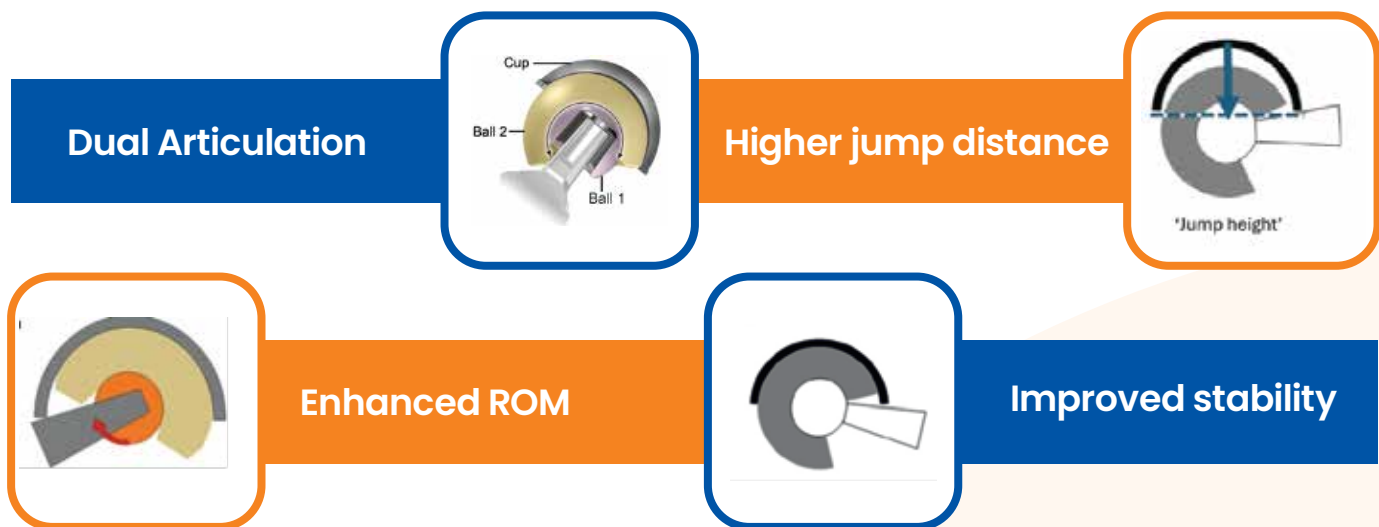
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Introduction

The **ETERNITY™** Dual Mobility Cup System is designed to provide enhanced joint stability, increased range of motion, and reduced risk of dislocation in primary and revision total hip arthroplasty. The dual mobility concept combines a highly polished femoral head, a mobile vitamin E liner, a shell-metal liner interface, and a porous acetabular shell to optimize articulation while preserving biomechanical function.

The system is intended to help surgeons achieve reliable fixation, restore hip biomechanics, and improve long-term clinical outcomes across a wide range of patient anatomies.

Design Objectives



Indications

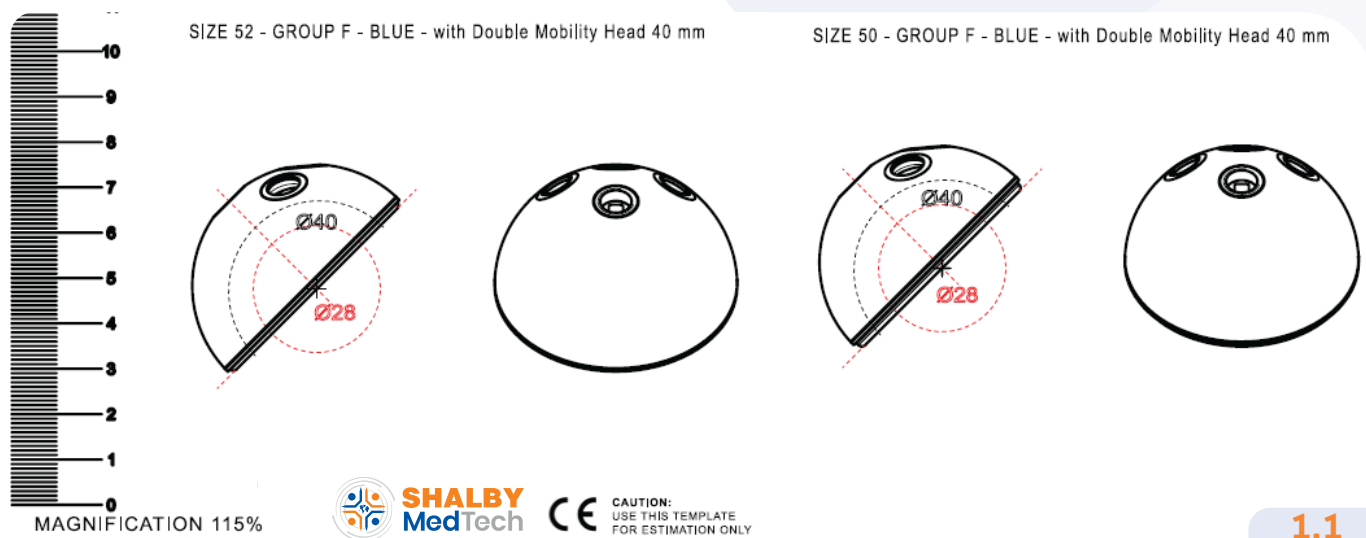
The **ETERNITY™** Dual Mobility Cup System is intended for use in primary and revision total hip arthroplasty where reconstruction of the hip joint is required.

It is indicated for patients with:

- Primary Osteoarthritis
- Rheumatoid Arthritis
- Post-traumatic Arthritis
- Avascular Necrosis
- Femoral Neck Fracture
- Failed Previous Hip Arthroplasty
- Revision Total Hip Replacement
- Hip Dysplasia (where appropriate)

Contraindications

- Local or systemic infection
- Septicaemia
- Persistent acute or chronic osteomyelitis
- Confirmed nerve or muscle lesion compromising hip joint function.



Preoperative Planning

The **ETERNITY™** Dual Mobility Cup System components should be implanted only by surgeons trained for joint replacement procedures. Preoperative planning provides useful information for the correct placement of the implant but does not necessarily indicate the appropriate cup size.

The correct cup size must be determined during surgery. To achieve the best results preoperative planning using special templates (with specific magnification always advisable). [Fig 1.1]

It's suggested to do AP radiograph with adequate contrast.

In order to achieve successful hip replacement surgery, it is crucial to plan the procedure pre operatively, taking into consideration the patient's individual anatomy and level of physical activity. The surgeon should conduct a thorough evaluation of the patient's clinical condition to determine the correct implant type and size, as well as its final intraosseous position.

Positioning of Patient and Surgical Approach

During **ETERNITY™** Dual Mobility hip replacement surgery, various surgical approaches can be utilized to implant the components. The following steps are applicable for, postero-lateral, anterior and other surgical access routes.

The patient is placed in a lateral position for the procedure. The incision is made postero-laterally, followed by opening of the fascia lata. The external rotator muscles are then resected, and the joint capsule is incised.

The femoral head is dislocated dorsally to allow for easy access and removal of the head from the socket. This is achieved by flexing the hip and abducting the leg, which allows the femoral head to dislocate freely.

These steps are critical for a successful hip replacement surgery, regardless of the surgical approach utilized.

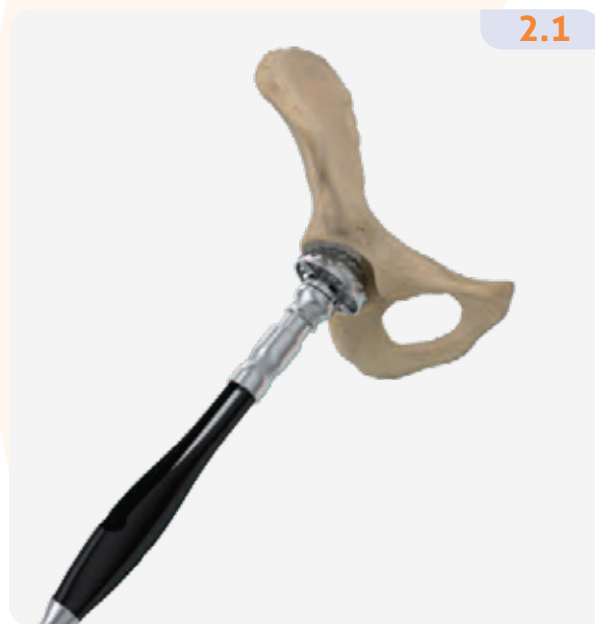
Acetabular Exposure

Before acetabular reaming, it is essential to have a clear and direct view of the acetabular site, regardless of the surgical approach selected. This requires the removal of any soft tissues and osteophytes that could obstruct visibility, allowing for a complete view of the entire acetabular socket. This is crucial to identify the presence of any cavitory or segmental defects and ensure accurate diagnosis and treatment. Specific acetabular retractors are needed to facilitate acetabular exposure.

Acetabular Bone Preparation

Begin preparation of the acetabulum using a reamer smaller than the acetabular **ETERNITY™** porous acetabular shell determined in the preoperative plan.

Increase the size of the acetabular drill each time (double diametrical increments). The last acetabular reamer must show bleeding cortical bone and highlight an optimal site in terms of shape, depth and bone support. A final check can be made by analyzing the bone left inside the acetabular drill itself. [Fig 2.1]



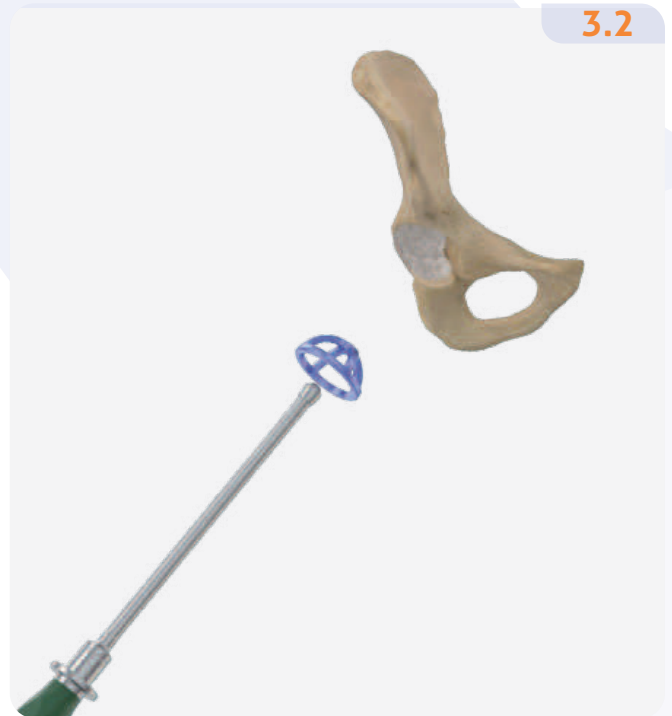
2.1

Trial Cup

Following the use of the last reamer, take the trial **ETERNITY™** porous acetabular shell of the same size as the last acetabular reamer used. Screw it onto the impactor [Fig 3.1] and check the congruity of the milled acetabular opening around the trial cup. [Fig 3.2]



3.1



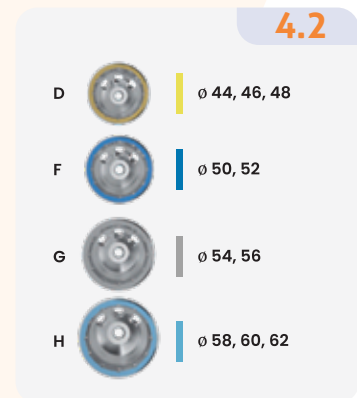
3.2

Equatorial Rim Color Code

The pictures shows the different color coding adopted to identify the different **ETERNITY™** porous acetabular shell internal taper and the coupling with liners. [Fig 4.1/4.2]



4.1



4.2

Compatibility Chart

4.3

Acetabular Cup System Compatibility	Acetabular Cup	TINBIN INLAY	VIT-E POLY Liner	Head
	44 mm	36 mm	36 mm	22 mm
	46 mm			
	48 mm			
	50 mm	40 mm	40 mm	28 mm
	52 mm			
	54 mm	42 mm	42 mm	28 mm
	56 mm			
	58 mm	44 mm	44 mm	28 mm
	60 mm			
	62 mm			

Alignment and Final Cup Insertion

Remove the polar cap using a hexagonal screwdriver. The cup's equatorial rim is color coded to match the trial liner color and size. [Fig 4.1/4.2]

To align the **ETERNITY™** porous acetabular shell with the correct implant position, screw the appropriate bars to the aligner tool, 1 in the 45° seat and 2 in the 10° seats. Magnetically mount the aligner to the impactor and then screw the shell on the impactor. [Fig 5.2]

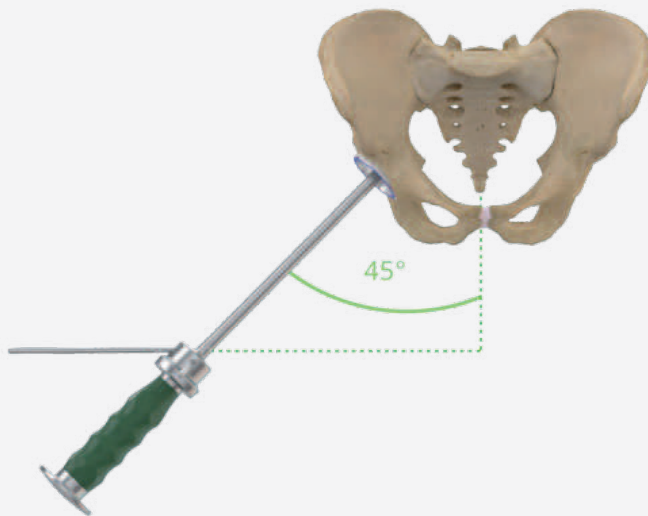
Then align the bar at 45° with the patient's sagittal plane (picture) and simultaneously one of the 2 bars at 10° perpendicularly with the patient's frontal plane. [Fig 5.3/5.4]



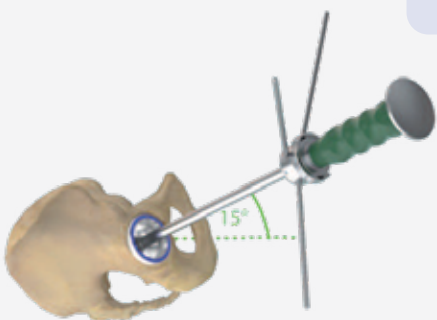
5.1



5.2



5.3



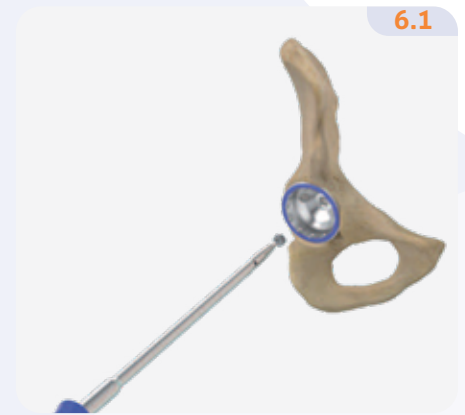
5.4



5.5

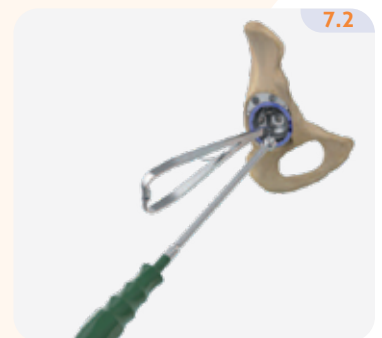
Polar Plug

Once the cup is being impacted, [Fig 5.5] proceed with the insertion of the polar cap [Fig 6.1]



Screw Application

In case of poor acetabular bone quality, the system allows the surgeon to further fix the cup utilizing bone screws. A hole is drilled into the bone with the help of the Drill Guide, which is inserted into the hole in the desired direction with a maximum angulation of approximately $\pm 15^\circ$. [Fig. 7.1] Use the Curved Depth Gauge to identify the correct length of the Bone Screw. Hold the screw with the Screw Clamp and insert it either with the Angled or Straight Screwdriver. [Fig. 7.2]



Trial Reduction with Trial Liner and Final Implant

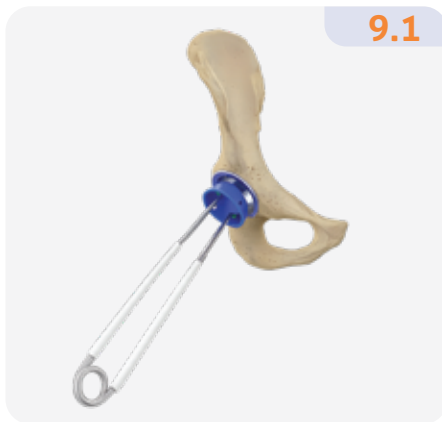
To carry out the trial reduction on the final implant, insert the trial liner matching the color code of the shell [Fig. 4.1/4.2], using a trial head compatible to the femoral head to be used for the implant. Attach the trial head on the femoral stem and reduce the hip. Check for joint stability and range of motion, making any necessary adjustments to restore joint mechanics. Make certain that prominent impinging bone and/or osteophytes are removed from the periphery of the acetabulum to maximize range of motion and stability. [Fig. 8.1 to 8.3]



Trial Reduction

Plug in the **ETERNITY™** Dual Mobility trial liner selecting the trial, whose color is equivalent to the color of the implanted shell rim. [fig 9.1] the size combinations can be found in the table [fig 4.3]

Plug in the trial Femoral head into the **ETERNITY™** Dual Mobility trial head, whose color is the same of the Dual mobility trial liner used previously, then proceed with the total joint trial reduction. [Fig 9.2/9.3]



9.1



9.2



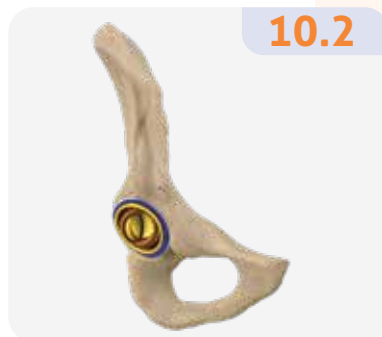
9.3

Liner Implantation

Proceed with inserting by hand in the final **ETERNITY™** Dual Mobility Liner and impact it with the standard liner impactor selecting the right impactor. [D32 for yellow Trial, and D36 for all other colors] [Fig 10.1 to 10.4]



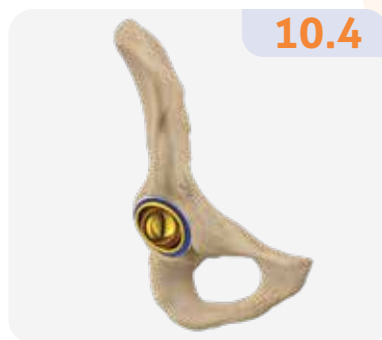
10.1



10.2



10.3



10.4

Press-Assembly

Back table assembly of the **ETERNITY™** vitamin E liner and corresponding Femoral Head is required. Proceed by assembling the **ETERNITY™** Dual Mobility Press component by placing the Press Stand flat on the table and screw on the T-Handle and the plastic cone portion of the press. Put the vitamin E liner Supporting Piece into the base of the Press. [Fig. 11.1 to 11.3]



Final Femoral Head and ETERNITY™ vitamin E liner Assembly

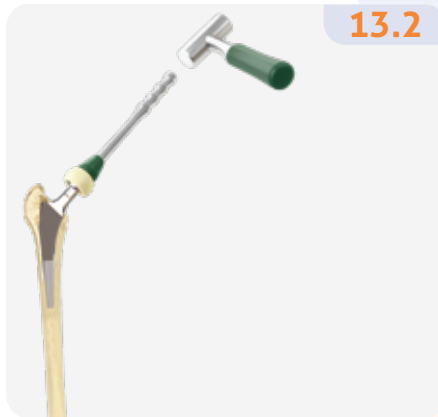
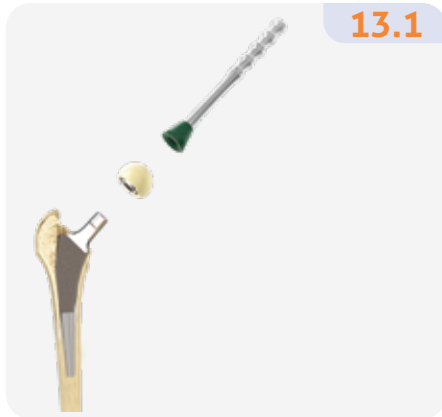
Open the press by turning the T-handle counter-clockwise. Place the vitamin E liner onto the Head Supporting Piece and then place the Femoral Head onto the vitamin E liner. Once the Femoral Head and vitamin E liner are in a vertical position, tighten the Press until they are fully lodged together. After insertion, the air confined between the head and insert is usually released, resulting in a characteristic noise which gradually releases the pressure on the T-handle. After the Femoral Head and vitamin E liner are assembled, verify that the coupling has complete mobility. [Fig. 12.1 to 12.3]

NOTE: Ensure that the inside of the shell metal liner is clean and free of soft tissue or other debris, which could prevent the vitamin E liner insert from properly seating in the shell metal liner.



Final Femoral Head and ETERNITY™ Vitamin E liner Implantation

Plug the final femoral head on the stem taper, making sure that the final stem taper is cleaned and does not present any tissue residuals. After positioning, gently impact the final Femoral Head and ETERNITY™ vitamin E liner with the standard femoral head impactor. Proceed with the final implant reduction. [Fig 13.1 to 13.3]



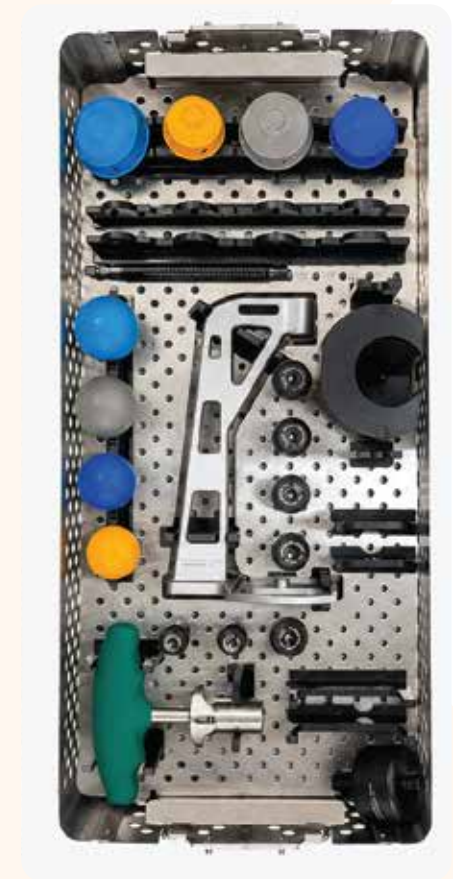
Instrument set for ETERNITY™



Tray-1



Tray-2



Tray-3

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For additional information, please contact your local Shalby MedTech distributor, or reach our customer service team at **+9163570 07339** or **infomedtech@shalby.org**.



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